4.4 Congruence and Transformations

Name _____ Period ____ Date _____

Geometry 4.4 (30-minute plan) · Student sheet

I can…

- describe a congruence transformation that maps one figure onto another.
- decide whether two figures are congruent.
- use the theorems about reflections in parallel and intersecting lines.

Vocabulary · fill in the missing word

Congruent figures — figures with the same size and _____

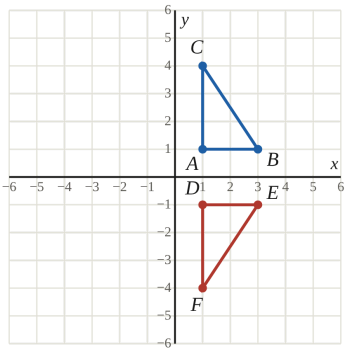
Congruence transformation — a transformation that _____ length and angle measure (a translation, reflection, rotation, or a composition of them)

Part 1 · Notes and examples (fill in each note, then work the example)

Congruence: two figures are congruent if and only if there is a _____ of rigid motions that maps one onto the other.

Example 1

Describe a congruence transformation that maps △ABC to △DEF: A(1, 1), B(3, 1), C(1, 4) and D(1, −1), E(3, −1), F(1, −4).



Example 2

Describe a congruence transformation that maps △ABC with A(−4, 1), B(−2, 1), C(−4, 3) onto △PQR with P(4, −3), Q(2, −3), R(4, −1).

Reflections in Parallel Lines Theorem: reflecting in two parallel lines is the same as a _____, twice the distance between the lines.

Reflections in Intersecting Lines Theorem: reflecting in two intersecting lines is the same as a _____ about their intersection by twice the angle between them.

Example 3

Reflect (0, 2) in the line x = 1, then reflect the image in x = 4. What single transformation does the same thing?

Part 2 · Partner practice

Partner practice

A figure is reflected in x = −1 and then in x = 2. Describe the single transformation.